

Day 98: Weekly Revision — Statistics, Probability and Calculus

1. Day number

Day 98

2. Topic name

Revision of Days 92–97: Statistics, Probability and Calculus

Today you will revise the mathematical ideas you learned before moving deeper into machine learning.

3. Connection

These topics form part of the **basic mathematical foundation for machine learning**.

You have learned three broad ideas:

Statistics

→ understand and summarize data

Probability

→ reason about uncertainty

Calculus

→ understand change and reduce model error

Together, they help answer questions such as:

- What does our dataset look like?
 - How uncertain is an outcome?
 - How spread out are the values?
 - How can a model improve its parameters?
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4. Revision summary of Days 92–97

Day 92 — Descriptive Statistics

You learned how to summarize numerical data.

Important ideas:

Mean

→ average

Median

→ middle value

Variance

→ squared measure of spread

Standard deviation → easier-to-interpret measure of spread

For example:

10, 20, 30

Mean:

$$(10 + 20 + 30) / 3 = 20$$

Median:

20

Day 93 — Probability Fundamentals

Probability describes **how likely an event is**.

For equally likely outcomes:

$$\text{Probability} = \frac{\text{favorable outcomes}}{\text{total possible outcomes}}$$

Probability ranges from:

0 → impossible

1 → certain

For example, on a fair six-sided die:

$$P(\text{rolling } 4) = 1/6$$

You also learned the complement:

$$P(\text{not } A) = 1 - P(A)$$

Day 94 — Basic Probability Distributions

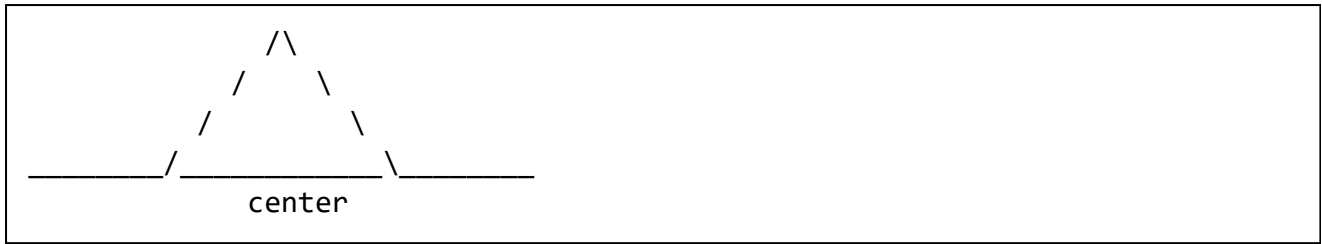
A **distribution** shows how values or probabilities are spread across possible values.

You learned:

Discrete
→ separate/countable possibilities

Continuous
→ values along a measurement range

You also met the **normal distribution**, whose general shape is:



In a bell-shaped distribution, values are more concentrated near the center and less concentrated far from it.

Day 95 — Central Limit Theorem

You learned about:

- population
- sample
- sample mean
- repeated samples
- sampling distribution

The main intuition was:

If we repeatedly take sufficiently large random samples and calculate their means, those means often form an approximately normal distribution around the population mean.

Conceptually:

```
Population
  ↓
sample → mean
sample → mean
sample → mean
sample → mean
  ↓
distribution of means
```

Day 96 — Derivatives

A derivative describes the **local rate of change** of a function.

You learned:

Positive derivative
→ function increasing locally

Negative derivative
→ function decreasing locally

Zero derivative
→ function locally flat

In machine learning, derivatives help us understand how the model's error changes when we adjust a parameter.

Day 97 — Gradients and Gradient Descent

The gradient tells us how loss changes with model parameters.

Gradient descent uses that information to move toward **lower loss**.

The simplified idea is:

```
parameter
=
parameter - learning_rate × gradient
```

The important intuition:

Gradient points uphill.

Gradient descent moves downhill.

5. Important topics

Your main revision vocabulary is:

Topic	Beginner meaning
Mean	Average value
Median	Middle value
Variance	Squared measure of spread
Standard deviation	Typical spread around the mean
Probability	Likelihood of an event
Distribution	How values/probabilities are spread
Normal distribution	Bell-shaped distribution
CLT	Behavior of means from repeated samples
Derivative	Local rate of change
Gradient	How loss changes with parameters
Gradient descent	Repeatedly changing parameters to reduce loss

You do not need advanced formulas yet. Make sure you understand **what each idea tells you**.

6. Foundational notes

These topics are connected rather than completely separate.

Imagine you have customer purchase data.

First, statistics can tell you:

What is average spending?
How spread out is spending?

Probability can help you think about:

How likely is a certain outcome?

Distributions can tell you:

Where do values tend to appear?

Sampling and the CLT help you understand:

What happens if we repeatedly estimate something using samples?

Finally, derivatives and gradients help when building models:

```
Current model
  ↓
calculate error
  ↓
find direction of change
  ↓
adjust parameters
  ↓
try to reduce error
```

7. Easy example

Consider:

2, 4, 6, 8, 10

Mean

Add everything:

$2 + 4 + 6 + 8 + 10 = 30$

Divide by 5:

Mean = $30 / 5$
= 6

Median

The values are already sorted:

2, 4, 6, 8, 10
 ↑

So:

Median = 6

Spread

Not every value equals 6.

Some are below it:

2, 4

Some are above it:

8, 10

So the data has some spread around its center.

Probability example

Imagine we randomly choose one value from the dataset.

The probability of choosing a value greater than 6 is based on:

favorable values:
8, 10

There are 2 favorable values out of 5 total:

$$P(\text{value} > 6) = 2/5 = 0.4 = 40\%$$

This small example connects descriptive statistics with probability.

8. Revision problem statement

Use this tiny dataset:

10, 20, 20, 30, 40

Complete these tasks:

1. Calculate the **mean**.
2. Find the **median**.
3. Describe whether the values have no spread, small spread, or noticeable spread.
4. If one value is chosen randomly, calculate the probability that it equals 20.
5. Explain what would happen conceptually if you repeatedly took small random samples and calculated their means.
6. Imagine these numbers were related to model error. Explain how a **gradient** could tell the model which direction to change a parameter to reduce that error.

Keep your explanation simple.

9. Concepts used

The revision problem combines:

```
Dataset
↓
Mean + Median
↓
Center

Dataset
↓
Variance / Standard deviation idea
↓
Spread

Possible outcomes
↓
Probability

Repeated samples
↓
Sample means
↓
Sampling distribution / CLT

Model error
↓
Derivative / Gradient
↓
Gradient descent
↓
Lower error
```

This is the mathematical bridge toward machine-learning models.

10. Thought process

When solving the revision exercise, separate it into small parts.

Part A — Understand the center

Ask:

| What value represents the dataset reasonably well?

Calculate the mean and median.

Part B — Understand the spread

Ask:

| Are all values close to the center, or are some far away?

You do not need a complicated variance calculation unless requested.

Part C — Think about probability

Identify:

favorable outcomes
total outcomes

Then use:

$\text{probability} = \text{favorable} / \text{total}$

Part D — Think about repeated samples

Ask:

If I repeatedly select samples and calculate their averages, where should most of those averages tend to appear?

Connect this to the CLT.

Part E — Connect to optimization

Imagine there is a parameter affecting model error.

Ask:

Does increasing the parameter increase error?
or
Does increasing the parameter decrease error?

The derivative or gradient helps answer this.

Then gradient descent moves toward lower error.

11. Beginner-friendly steps

For the dataset:

10, 20, 20, 30, 40

Follow this sequence:

STEP 1

Count the values.

STEP 2

Add all values.

STEP 3

Divide by the number of values.

→ mean

STEP 4

Sort the values if necessary.

STEP 5

Find the middle.

→ median

STEP 6

Compare values with the center.

→ think about spread

STEP 7

Choose the event:

"value equals 20"

STEP 8

Count how many 20s exist.

STEP 9

Use:

favorable / total

→ probability

STEP 10

Imagine repeatedly sampling the data.

STEP 11

Think about where the sample means
would tend to gather.

STEP 12

Imagine a model parameter controlling error.

STEP 13

Use the gradient to decide
which direction may lower error.

STEP 14

Take a small step and repeat.

12. Suggested solving approach

Use a **conceptual + simple calculations** approach.

Do not mix everything together immediately.

A good order is:

1. Describe the dataset
2. Calculate its center
 - mean
 - median
3. Discuss its spread
 - values near/far from mean
4. Solve one simple probability
5. Discuss repeated samples
 - CLT idea
6. Connect derivative/gradient to model error
7. Explain gradient descent using small repeated updates

At this stage, understanding the **meaning** of the mathematics is more important than performing long calculations.

13. Easy edge cases

All values are equal

5, 5, 5, 5, 5

Then:

Mean = 5
Median = 5
Variance = 0
Standard deviation = 0

There is no spread.

One value

10

The mean and median are both 10.

There is no spread in the simple population-style interpretation.

Impossible probability

If the possible values are:

1, 2, 3

then:

$P(\text{selecting } 10) = 0$

Certain probability

For the same values:

$P(\text{selecting a value} \leq 3) = 1$

Gradient is zero

If:

$\text{gradient} \approx 0$

the error surface may be locally flat.

The model might already be near a minimum, although later you will learn that zero gradient does not always guarantee the best possible point.

Learning rate too large

The model may:

overshoot
→ jump past the low-error region

Learning rate too small

The model may:

move correctly
→ but very slowly

14. Common mistakes to avoid

- Do not confuse **mean** with **median**. Mean uses every value; median is based on the middle position after sorting.
 - Do not think standard deviation represents the center. It represents **spread**.
 - Do not write probabilities outside 0 to 1; percentages should stay between 0% and 100%.
 - Do not assume every dataset has a normal distribution. The normal distribution is one important possible shape.
 - Do not confuse the distribution of original values with the **sampling distribution of sample means** in the CLT.
 - Do not think a derivative only means "increase." It can be positive, negative, or zero.
 - Do not think gradient descent automatically jumps straight to the best answer. It normally uses **repeated updates**.
 - Do not forget the learning rate: direction and step size both matter.
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15. Quick self-check questions

1. What is the difference between **mean** and **median**?
2. What does a large standard deviation generally tell you about the dataset?
3. If an event has probability 0.25, what percentage probability is that?
4. In simple terms, what happens to the distribution of sample means under the Central Limit Theorem?
5. If the loss gradient is positive for one parameter, in which general direction does gradient descent adjust that parameter?

Try answering each question in **one or two sentences**, rather than memorizing a definition.

16. Hint only

For the revision dataset:

10, 20, 20, 30, 40

For the mean, begin with:

$10 + 20 + 20 + 30 + 40$

Then divide by:

5

For the median, look at the **middle value**.

For:

$P(\text{value} = 20)$

count:

How many 20s?

How many total values?

For the CLT part, imagine producing:

sample 1 → average

sample 2 → average

sample 3 → average

...

and ask where those averages would tend to gather.

Finally, for gradient descent, remember:

Gradient

→ tells you how error changes.

Gradient descent

→ takes small steps in the direction
that aims to reduce that error.

The central revision picture is:

DATA

↓

Statistics

→ What does the data look like?

UNCERTAINTY

↓

Probability + distributions

→ What outcomes are possible and likely?

SAMPLING

↓

CLT

→ How do repeated sample averages behave?

MODEL ERROR

↓

Derivative + gradient

→ Which direction changes error?

OPTIMIZATION

↓

Gradient descent

→ Repeatedly move toward lower error

That is the mathematical foundation you will begin applying when you move into actual machine-learning models.